

Title:

The health and economic value of Artificial Intelligence in a resource-constrained diagnostic pathway of prostate cancer in the Netherlands – a discrete event simulation model

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Appendix 2

This appendix provides additional information on the input parameters used in the model, covering:

- Extrapolation of prostate cancer incidence
- Diagnostic performance data
- Extrapolated number of individuals entering the diagnostic pathway
- AI impact on diagnostic accuracy

Extrapolation of prostate cancer incidence

The increase in the number of prostate cancer patients was based on the increasing number reported by the Netherlands Cancer Registry(NCR) [61]. The NCR is a registry that facilitates healthcare research and tracks cancer incidence in The Netherlands. In the NCR, only the first incidence of prostate cancer is registered, in line with international agreements to allow the comparison between regions. All age groups were considered in this study because all add to the demand in the prostate cancer detection pathway. The numbers were linearly extrapolated in R based on the reported numbers between 1989 and 2024 (see Figure S1). The yearly increase in the number of prostate cancer patients was estimated at 281 patients per year.

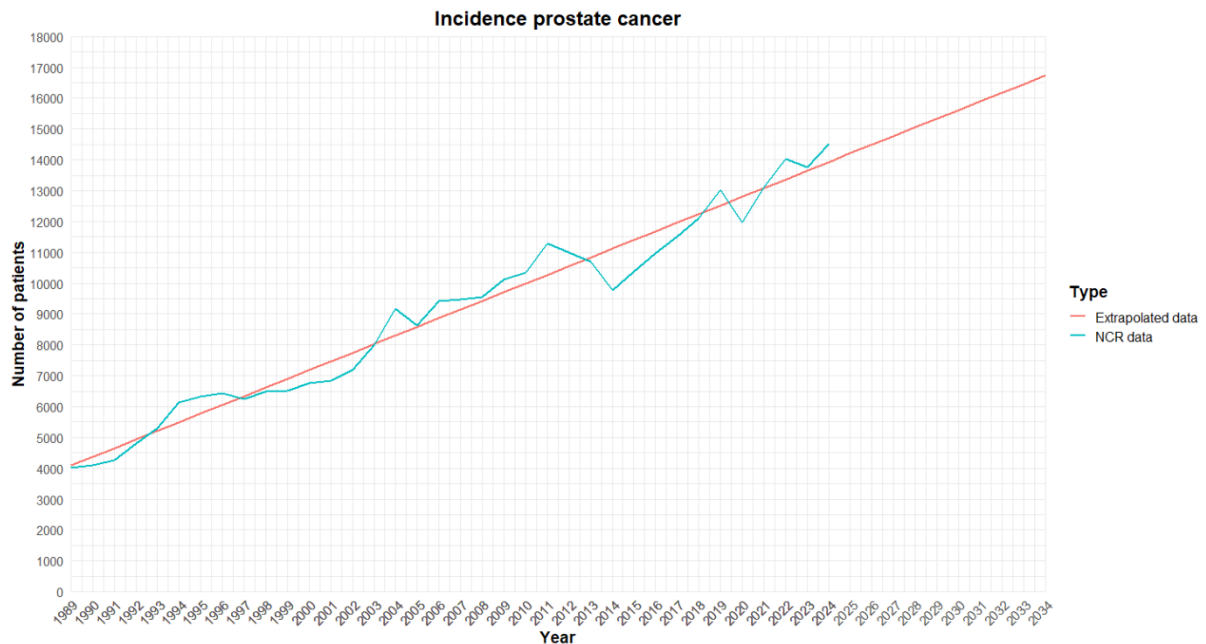


Figure S1. Projected growth of prostate cancer incidence.

The number of individuals that entered the pathway is a multiplication of the extrapolated patients by the diagnostic performance (including the PSA, DRE, MRI assessment, and biopsy assessment) of detecting significant and insignificant/low-risk PCa combined, using the formula:

$$(1) \text{ \#undiagnosed patients} = \text{\#diagnosed patients} / (\text{sensitivity PSA} * \text{sensitivity urologist} * \text{sensitivity MRI assessment} * \text{sensitivity pathologist} * \text{incidence})$$

in which the *\#diagnosed patients* are the number of patients extrapolated and the *incidence* the incidence of prostate cancer, which equals 11.1% [70,71].

Diagnostic performance data

Table S1 provides an overview of the diagnostic performance for prostate cancer detection used in this simulation study. The sensitivity was used to determine the test result for PCa patients (true positive or false negative), the specificity was used to determine the result for individuals without PCa (true negative or false positive).

For the PSA + DRE, the sensitivity for significant PCa was based on the combined sensitivity of PSA and DRE separately, using the following formula:

$$(2) \text{ Sensitivity PSA + DRE} = \text{PSA sensitivity} + \text{DRE sensitivity} - (\text{PSA sensitivity} * \text{DRE sensitivity})$$

The specificity of PSA + DRE was based on the combined specificity of PSA and DRE with a negative outcome only in case of both tests being negative. The following formula was used:

$$(3) \text{ Specificity PSA + DRE} = \text{PSA Specificity} * \text{DRE Specificity}$$

Table S1. Overview of the diagnostic input parameters used in the base case.

Test	Performance		Value	Reference
PSA + DRE <i>(cut-off > 3 ng/ml)</i>	Significant*	Sensitivity	0.98	[368–370]
	Insignificant/ low-risk**	Sensitivity	0.46	[371]
	Combined	Sensitivity	0.99	Calculated
	No PCa	Specificity	0.16	[372]
	Significant	Sensitivity	0.65	[373]
	Insignificant/	Sensitivity	0.70	[373]

Urologists	low-risk			
	Combined	Sensitivity	0.90	Calculated
	No PCa	Specificity	0.61	[373]
MRI / Radiologists	Significant	Sensitivity	0.91	[54]
	Insignificant/ low-risk	Sensitivity	0.70	[54]
	Combined	Sensitivity	0.97	Calculated
	No PCa	Specificity	0.58	[57]
	Significant	Sensitivity	0.80	[54]
Biopsy / Pathologists	Insignificant/ low-risk	Sensitivity	0.51	[54]
	Combined	Sensitivity	0.90	Calculated
	No PCa	Specificity	1.00	[54]

*Estimated based on the sensitivity of PSA combined with the sensitivity of DRE

**The sensitivity and specificity of any prostate cancer were considered.

Extrapolated number of individuals entering the diagnostic pathway

The extrapolated numbers and the estimated number of individuals entering the pathway are reported in Table S2.

Table S2. Overview of the number of prostate cancer patients that enter the diagnostic pathway each year.

Incidence year	Number of patients extrapolated	Number of patients that enter the pathway
2025	14,211	164,592
2026	14,492	167,847
2027	14,773	171,102
2028	15,054	174,356
2029	15,335	177,611
2030	15,617	180,877

2031	15,898	184,131
2032	16,179	187,386
2033	16,460	190,640
2034	16,741	193,895

AI impact on diagnostic accuracy

AI-MRI

In the AI-MRI strategy, an AI-accelerated MRI protocol is simulated. Based on the literature, the acceleration of MRI can be performed using multiple methods [59,73]. The fundamentals of MRI acceleration are based on the reduced acquisition of data which also results in loss of diagnostic information, although often non-inferior [59,72,73]. In the current study, the downward trend of diagnostic information was simulated by reducing the diagnostic performance versus the performance used in the base case by 2% [74]. Table S3 shows the performance of the MRI assessment for a 3-time acceleration (R3). The R3 was simulated in the model by reducing the scan time from 25 minutes to 8.33 minutes.

Table S3. Overview of the MRI performance characteristics used in the AI-MRI strategy.

Performance		Base case	Percentage difference	AI-MRI
Significant PCa	Sensitivity	0.91	-2	0.89
Insignificant/low-risk PCa	Sensitivity	0.70	-2	0.69
No PCa	Specificity	0.58	-2	0.57

AI-assessment

In the AI-assessment strategy, an AI-assisted prostate cancer diagnosis is simulated. Multiple methods have been described of how AI can assist a radiologist in its diagnosis, ranging from highlighting suspicious areas to autonomous detection for certain cases [55,57]. Within the current study, the AI-assessment strategy is based on the study of Twilt et al. (2026)[74] which reported an specificity of 0.692 (an improvement in the specificity of 11.5 percentage points) while maintaining the sensitivity equal to the base case and reducing the time required by the radiologist by half (Table S4).

Table S4. Overview of the MRI performance characteristics used in the AI-assessment strategy.

Performance		Base case	Percentage difference	AI-assessment
Significant PCa	Sensitivity	0.91		0.91
Insignificant/low-risk PCa	Sensitivity	0.70		0.70
No PCa	Specificity	0.58	+11.5*	0.69

*percentage point difference

AI-MRI and AI-assessment

In the third strategy, the AI-MRI application and the AI-assessment application are combined. The resulting MRI performance characteristics are provided in Table S5.

Table S5. Overview of the MRI performance characteristics used in the combined AI-MRI & AI-assessment strategy.

Performance		Base case	Percentage difference	AI-MRI and AI-assessment
Significant PCa	Sensitivity	0.91	-2	0.89
Insignificant/low-risk PCa	Sensitivity	0.70	-2	0.69
No PCa	Specificity	0.58	-2, +11.5*	0.68

*percentage point difference